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Research Article

Developing a User-Centric Quality Model for Gaming as a Service (GaaS): Enhancing User Satisfaction Through Key Quality Factors

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This study presents a comprehensive and user-centric quality model for gaming as a service (GaaS), grounded in a multistage survey methodology involving pretest, postgame, and posttest evaluations. The research identifies and empirically validates key quality attributes that influence user satisfaction, including controllability, responsiveness, accessibility, cost transparency, security, and social features. Data from 62 cloud gamers, analyzed through ANOVA and regression techniques, reveal that users prioritize high-resolution graphics, diverse game libraries, intuitive controls (ICs), and seamless audio–visual performance. The findings highlight a strong alignment between user expectations and the proposed quality model. Practical recommendations are offered for GaaS providers, focusing on improved user onboarding, transparent system requirements, enhanced social features, and robust security protocols. The study also discusses emerging technologies such as AI-driven personalization and adaptive streaming, which hold promise for enhancing quality of experience (QoE) in dynamic network conditions. Future research should include larger and more diverse user samples, longitudinal analysis, and cross-cultural perspectives to further validate and refine the model.

Keywords: adaptive streaming; AI; cloud gaming; gaming as a service; quality of experience; survey-based research; user-centric model; user satisfaction

1. Introduction

The video game industry has evolved into one of the most profitable entertainment sectors, surpassing both the film industry and sports in terms of revenue [1]. A notable shift in this domain is gaming as a service (GaaS), a cloud-based model that allows users to stream high-quality video games without the need for expensive hardware [2]. GaaS has the potential to make gaming more accessible and inclusive, especially in an era where the demand for gaming experiences continues to rise among diverse user groups [3]. This model aligns with earlier trends observed in the gaming industry, such as the emergence of online multiplayer in the early 2000s, which significantly changed how players engage with games [4]. However, the success of GaaS platforms depends on their ability to deliver high-quality service that meets user expectations consistently [5]. Research by Roettl et al. [6] highlights that GaaS still faces challenges in meeting these expectations, particularly regarding streaming stability and device compatibility.

Despite the promise of GaaS, several challenges remain, such as instability in streaming and unclear pricing models [7]. These challenges indicate a gap between the current offerings of GaaS platforms and user expectations. This issue is echoed in the work of Shrivastava and Damle [8], who emphasized the importance of performance stability for long-term user satisfaction. To address these concerns, this article proposes a user-centric quality model for GaaS, building on the findings of Moina-Rivera et al. [9], who identified the lack of clear user satisfaction metrics as a major barrier to improving

cloud gaming services. Through a combination of survey insights and an extensive literature review, this research aims to identify the key factors influencing user satisfaction and to provide a practical framework that helps GaaS providers align their offerings with user needs.

The key contributions of this article are:

- Development of a user-centric quality model that identifies critical quality factors for GaaS platforms, emphasizing user satisfaction, which is aligned with existing models in cloud computing [10].
- A survey-based empirical analysis to identify and validate key quality factors that most significantly influence user experience, extending findings from previous research [11].
- A practical framework for GaaS providers with actionable recommendations to improve service features like streaming stability, access, and payment systems, addressing the gaps noted by previous studies [7].
- Identification of key challenges in GaaS, such as latency and service interruptions, that hinder user satisfaction, providing a clear basis for future improvements as proposed by Moina-Rivera et al. [9].
- Advancement of GaaS research by offering deeper insights into user needs and expectations, building on the foundations of quality of experience (QoE) models in cloud gaming [5].

To develop this user-centric quality model, we conducted a comprehensive survey targeting GaaS users to gather insights into the quality aspects that matter most to them. These findings, combined with our literature review, highlighted critical factors such as streaming stability, ease of access, payment simplicity, and service responsiveness concerns that align with those raised by Shrivastava and Damle [8]. The proposed model is intended to serve as a practical framework for GaaS providers, guiding them in enhancing their services to better meet user needs. Ultimately, our goal is to help providers deliver experiences that not only meet but exceed user expectations, fostering long-term engagement and satisfaction. This aligns with previous research emphasizing the importance of maintaining high service standards to ensure user retention [12].

1.1. Evolution of Gaming Consoles and the Emergence of GaaS. Since the release of the first home video game console, the Magnavox Odyssey, in 1972, the gaming industry has undergone significant technological advancements [13]. Early innovations—such as ColecoVision's 16-bit graphics and the integration of hard disk technology during the 1987 rivalry between the Sega Genesis and TurboGrafx-16 [14]—marked the beginning of this evolution. These developments were followed by the emergence of 3D graphics, introduced in games like Super Mario 64 and Final Fantasy VII, highlighting the industry's ongoing drive for technological advancement [15].

The dominance of the PlayStation 2 in the early 2000s, followed by the rise of the PlayStation 3, Xbox 360, and Nintendo Wii, marked a new era of high-definition graphics and online multiplayer gaming—signaling a shift toward more immersive and socially connected experiences [4]. As hardware costs escalated and traditional gaming models became increasingly restrictive, the industry saw the emergence of GaaS in the 2010s. This model enabled users to stream games directly to their devices, eliminating the need for high-end hardware [3]. The shift to GaaS aligns with earlier trends in gaming, where advances in networking and computing technologies consistently reshaped the gaming landscape [16].

However, while GaaS offers a more affordable and accessible alternative to traditional gaming setups, several challenges persist, including streaming quality, access limitations, and the need for more flexible payment systems [8]. The growing demand for high-quality gaming experiences on average devices further underscores the need for innovation in GaaS platforms [6]. Despite recent advancements, gaps remain in service stability and reliability, both of which are crucial for enhancing user satisfaction [7]. This article aims to identify and address these gaps by proposing a user-centric quality model for GaaS platforms, designed to better meet user needs and improve the overall user experience.

2. Literature Review

The gaming industry is undergoing a profound transformation with the rise of GaaS, which enables access to cloud-based gaming platforms [17]. Within GaaS environments, the quality of the gaming experience is a critical factor that drives both user satisfaction and platform success [18]. This literature review explores existing research and quality models relevant to GaaS, aiming to identify the key elements that influence user satisfaction and to lay the groundwork for the development of a comprehensive and user-centric quality model.

Cloud computing quality models offer a foundational framework for evaluating the performance and reliability of cloud-based services [19]. While traditional models such as ISO/IEC 9126 have been adapted for cloud environments, the unique characteristics of GaaS demand more specialized frameworks [10]. GaaS platforms require models that account for the dynamic and interactive nature of gaming experiences, including gameplay quality, immersion, and user enjoyment [20]. Recent studies have underscored the challenges of assessing these factors—particularly with respect to latency, frame rate, and server availability—which are critical to overall user satisfaction [7, 9].

The integration of QoE metrics into GaaS frameworks is essential, as QoE captures the subjective dimensions of gaming, reflecting both user satisfaction and perception [5, 21]. Existing research highlights the complexity of evaluating quality in GaaS platforms, underscoring the need for models that incorporate both objective performance indicators and subjective user experiences [22]. Specialized frameworks are, therefore, necessary to address the multifaceted nature of GaaS quality, including factors such as latency, frame rate, and network stability [11]. Additionally, research by Shin [23, 24] on algorithmic mediation in online environments provides valuable insights into how user perceptions of QoE in cloud gaming may evolve with the introduction of AI-driven personalization and

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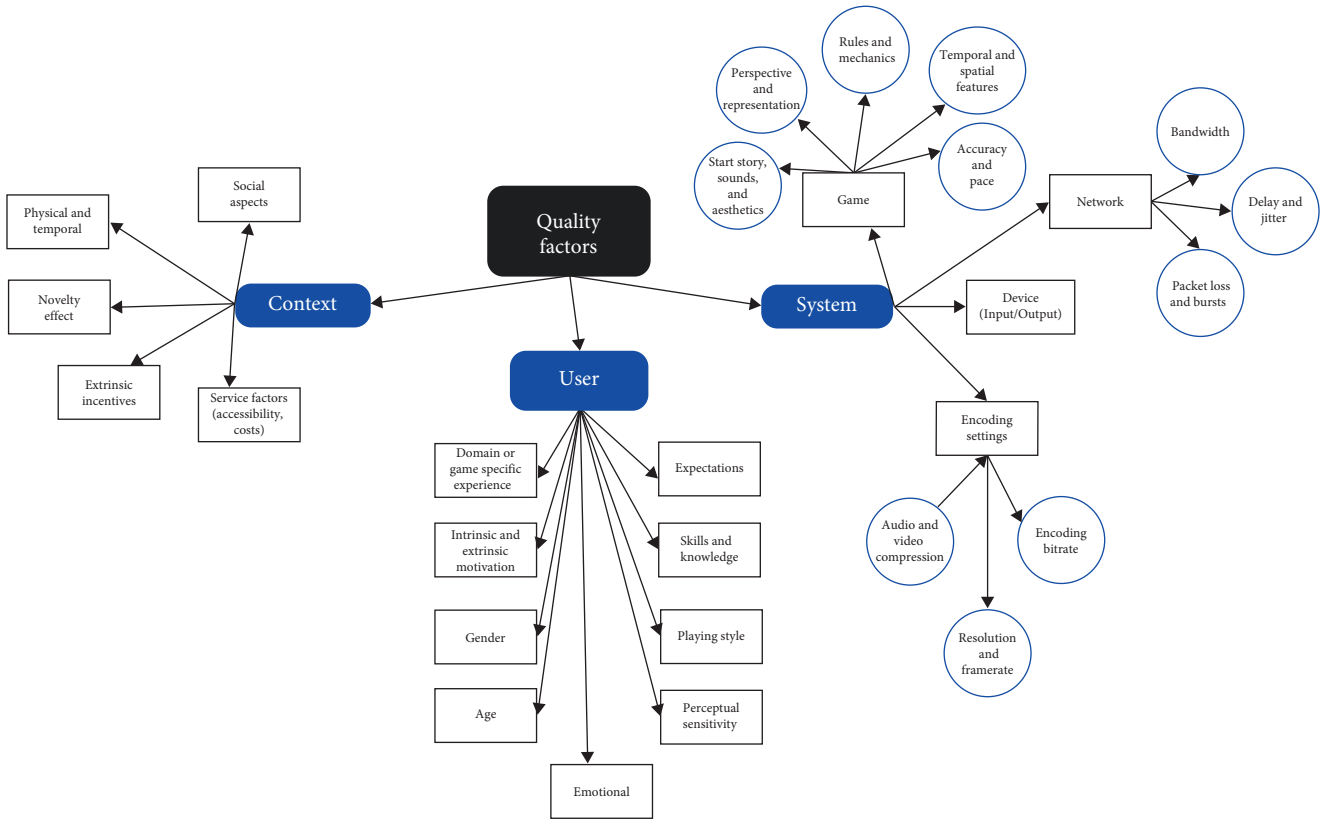

FIGURE 1: Overview of quality factors on gaming QoE.

recommendation systems, developments that can further inform QoE modeling in GaaS.

In summary, existing research underscores the need for specialized quality models to evaluate and enhance gaming experiences in cloud-based environments. Such models must integrate insights from both technical performance metrics and subjective user experiences, providing a more holistic approach to QoE assessment.

2.1. Influencing Factors on Gaming QoE. The previous section described the technical sophistication of cloud gaming systems. The factors influencing gaming QoE encompass several crucial dimensions that help us understand the complex dynamics of player experience. It begins with the technical infrastructure—hardware performance and network quality—which forms the foundation of gameplay quality.

Complementing this are game design elements, including mechanics, visuals, and narrative complexity, which collectively contribute to player immersion and engagement [25]. User interaction factors, such as input devices and interface design, significantly affect how players perceive and interact with the game environment [26]. Social dynamics within the gaming community—including multiplayer interactions and social platforms—also play a vital role in enhancing enjoyment and the longevity of the gaming experience [27]. Additionally, environmental and personal factors, ranging from individual preferences to ambient conditions, further shape the subjective quality of gameplay [28]. Considering these interrelated factors enables researchers and developers to optimize gaming experiences, ultimately boosting player satisfaction and engagement across diverse contexts.

Between 2013 and 2016, ITU-T Study Group 12 initiated the work item called G.QoE-Gaming [11], which defines gaming QoE as a multidimensional construct capturing the subjective perceptions and satisfaction levels of players [29]. This construct includes technical factors such as hardware performance and network stability, game design elements like graphics, mechanics, and narrative as well as social dynamics within gaming communities [12].

User interaction elements—input devices and interface design—also play a significant role. Environmental and personal factors, such as ambient conditions and individual preferences, further influence the overall perception of gaming quality. Understanding and optimizing G.QoE, therefore, requires a holistic approach [30]. Figure 1 provides an overview of the quality factors influencing gaming QoE, based on Laghari et al. [31].

Analyzing individual parameters offers a manageable approach to addressing the complex issue of gaming experience quality; however, the human element remains crucial. Player characteristics—such as experience, skills, and expectations—serve as key reference points that shape users' perceptions of quality [32]. Studies show that experienced gamers tend to be more sensitive to technical issues like low bit rate, while new gamers may not notice or find such issues disruptive [31]. This underscores the importance of incorporating user factors in QoE research.

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In addition, gaming experiences enhance skills and shape players' expectations, determining the expected level of quality. Players are categorized as "Novice" or "Expert" based on their experience and playing time. According to Vlahovic et al. [33], experienced players rank low-bit rate circumstances lower, but beginners rate them higher. This underscores how gaming expertise influences QoE evaluations. Similar findings in traditional online gaming studies validate the impact of previous gaming experiences on QoE perceptions under varied network conditions. In essence, gaming backgrounds significantly shape player expectations and assessments of gaming quality, indicating a crucial interplay between experience and perception in gaming contexts [11].

Current understanding of the contextual factors that influence gaming QoE is currently limited [34]. These factors summarize the situational aspects that characterize the user environment, including service costs and novelty effects, which pose challenges to comprehensive investigation within laboratory settings [11]. Although not the primary focus of the study, they are still important considerations for in-person user studies, which necessitate maintaining consistent lighting and noise conditions for a user-friendly environment that achieves optimal psychological comfort and results that mimic accurate user responses without stress [35].

Regarding social dynamics, research by Trepanowski et al. [36] shows that female players experience higher levels of stress and lower self-evaluations of their gaming skills when facing male opponents.

2.1.1. System Influencing Factors on Gaming QoE. The significance of system influencing factors on gaming QoE is paramount, especially regarding the impact of end-to-end delay [37]. Extensive research in both traditional online gaming and cloud gaming has delved into network and encoding settings, with end-to-end delay emerging as a dominant factor shaping gaming QoE. Studies by Jarschel et al. [38] and Sackl et al. [39] identified delay as a crucial influencer, although with varying effects across different game genres. Additionally, Raaen et al. [40] investigation unveiled the human perceptual system's sensitivity to motor–visual delays, revealing participants' ability to perceive delays as short as 40 ms.

Researchers [41] work demonstrated the mediating effect of gaming performance on the impact of delay, emphasizing the role of consistent performance in mitigating QoE degradation. Furthermore, Clincy and Wilgor [42] highlighted the substantial impact of packet loss on player experience, especially at rates of 1%, rendering games nearly unplayable.

In cloud gaming, the direction of packet loss is particularly significant compared to conventional gaming because it affects the video stream more than the control stream [43]. Research by Slivar et al. [44] highlighted the intensified impact of delay and packet loss on cloud gaming, showing a notably stronger effect on QoE than in traditional online gaming. They demonstrated an approximately linear relationship between packet loss and QoE, underscoring the critical importance of addressing both transmission direction and packet loss dynamics to optimize QoE and enhance player satisfaction in cloud gaming environments.

2.1.2. Encoding Parameters and Device Characteristics. The impact of encoding parameters, specifically bit rate and frame rate, on gaming QoE is a complex yet crucial aspect of gaming technology [45]. Synthesizing findings from various studies reveals the intricate relationship between these parameters and users' overall perception of gaming quality. Numerous investigations have explored the influence of bit rate on gaming QoE, with studies like Laghari et al. [46] showing that higher bit rates consistently enhance gaming experiences. However, Slivar et al. [44] caution that increasing frame rate at fixed bit rates may not always improve overall QoE, highlighting the need to optimize bit rates within bandwidth constraints to ensure the best gaming experience.

Moreover, the frame rate emerges as a critical encoding parameter affecting gaming QoE, as demonstrated by studies such as [47], which showcase significant differences in performance and quality ratings across varying frame rates. Additionally, the interaction effects between bit rate and frame rate add complexity, as demonstrated by Slivar et al. [44], emphasizing the importance of understanding these dynamics to maximize gaming QoE while minimizing bandwidth consumption.

2.1.3. Device Characteristics and QoE. Device characteristics significantly influence gaming QoE, yet their impact is subject to specific thresholds and evolving technological trends. Beyer et al. [47] examined gaming QoE across various screen sizes, noting an acceptance threshold between 3.27" and 5", beyond which QoE enhancements plateau. Huang et al. [48] conducted cloud gaming experiments across mobile and desktop platforms, revealing differing satisfaction levels concerning graphics and control quality.

Recent advancements in output devices, such as increased display resolutions, adaptive refresh rates (e.g., Nvidia's G-Sync and AMD's FreeSync), and enhanced color representation via HDR monitors, enable new rendering techniques like ray tracing to reach their full potential [49]. These developments underscore the evolving landscape of device characteristics and their dynamic role in shaping gaming QoE.

In the majority of studies investigating the quality factors impacting cloud gaming experiences, researchers typically focus on analyzing individual elements to ascertain their influence [50]. Each study often delves into one or two specific factors, aiming to demonstrate their predominant impact on the overall gaming experience. This approach allows researchers to examine the nuanced relationship between various quality factors and the user's perception of gaming quality [51]. By isolating specific elements such as network latency, video encoding, or input lag, researchers can gain deeper insights into their individual effects on the overall cloud gaming experience. However, it is essential to consider the interconnected nature of these factors and how they collectively contribute to shaping the user's experience in cloud gaming environments.

Multiple validated instruments such as GEQ, PENS, PXI, and ITC-SOPI have been used in gaming QoE studies. Appendix A outlines their key metrics and relevance to our research.

3. Methodology and Procedures

To develop a comprehensive and user-centric quality model for cloud gaming, a survey-based methodology was employed to

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Table 1: Pretest questionnaire [39].

Question	Answer options
1. What is your year of birth?	(Enter year)
2. What is your gender?	a. Female b. Male
3. Approximately how much time per week do you dedicate to gaming activities?	a. 0–1 h b. 1–5 h c. 5–10 h d. 10–20 h e. More than 20 h
4. How frequently do you engage in gaming sessions throughout the week?	a. Never b. 1–3 times a week c. 3–7 times a week d. 7–14 times a week e. More than 14 times a week
5. How would you characterize your level of gaming experience or proficiency?	a. Beginner b. Intermediate. c. Expert
6. On which kind of device do you usually play games?	a. PC (desktop) b. Smartphone/tablet c. Console (PlayStation, Xbox, etc.) d. Other
7. How experienced are you in playing the game "RIDE 4"?	a. Unexperienced b. Intermediate c. Expert
8. How experienced are you in playing the game "American Fugitive/GTA"	a. Unexperienced b. Intermediate c. Expert

assess key factors that influence user experience. This section outlines the research design, including the development of survey instruments, the testing environment, and the stages of data collection. By employing a structured approach, this study gathers insights across various stages of user interaction with cloud gaming platforms, from initial expectations to postsession evaluations. The methodology aims to address the critical factors that shape QoE in cloud gaming, ensuring a robust understanding of user perceptions and satisfaction.

3.1. Survey Design and Implementation. This research aims to develop a comprehensive user-centric quality model for cloud gaming by identifying key factors that influence user experience. To collect data on user perceptions of cloud gaming services, a survey-based methodology was employed. The survey captured insights across various dimensions, including cloud performance, network quality, and overall user satisfaction. The data were gathered at three distinct stages: pretest, postgame, and posttest. These stages were designed to capture the users' expectations before playing, their immediate experience during gameplay, and their reflections after the game session.

3.2. Pretest Questionnaire Development. The pretest questionnaire was designed to capture participant demographics and gaming habits, ensuring a broad representation of ages, experience levels, genders, and device types. A total of 62 respondents were surveyed, to include an equal number of males and females aged 18–34 years, as this demographic comprises a significant portion of the global gaming population [52].

As shown in Table 1, the questionnaire consisted of questions aimed at understanding gaming habits (e.g., frequency of gaming and time spent per week), level of experience (novice, intermediate, and expert), and the types of devices used (PC, smartphone, and console). Additionally, participants were asked about their experience with specific games (e.g., "RIDE 4," "American Fugitive," and "Grand Theft Auto (GTA)"). This approach helped ensure a representative sample that could provide reliable data across different player profiles.

3.3. Set-Up Test Environment. A methodical approach was adopted to establish a testing environment optimized for evaluating cloud gaming services. The testing was conducted on a dedicated gaming platform (Blacknut), which is the primary cloud gaming service available in Jordan. Given that cloud gaming platforms are still emerging in many regions, particularly in the Middle East, Blacknut was chosen due to its accessibility through Orange, a major telecommunications provider in Jordan [53]. This ensures the chosen platform is representative of current and accessible services in the region.

The testing environment involved using five different devices, including a PC, tablet, smartphone, and gaming console to ensure comprehensive coverage of different hardware configurations. A 4G internet connection was utilized to guarantee stable and uninterrupted connectivity during the test, which is critical for cloud gaming.

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Table 2: Postgame questionnaire.

Quality factor	Question	References
1. Appeal	• How important are cloud gaming platforms in offering games with high-quality graphics. • It is essential that cloud gaming platforms provide gaming accessibility and smooth performance comparable to PC gaming. • It is imperative that cloud gaming platforms offer a variety of game titles and subscription options to enhance their appeal to users.	[2]
2. Performance indication	• I should be able to easily evaluate my in-game performance. • I should be able to monitor and control the quality of image, audio, and video streaming. • The cloud gaming platform should maintain consistent performance overtime, avoiding issues such as stuttering, latency, or disconnections.	[54]
3. Intuitive controls	• It is important for cloud gaming platforms to offer games with controls and interfaces that are easy to remember. • The player's favorite games should be classified in a special part of the screen for easy access. • The game controls are intuitive.	[55]
4. Learnability	• It should be easy for me to become proficient with the game, and easy to operate and access. • Comments and prompts for all games in the library are provided to users to help them learn game mechanics, objectives, and strategies. • It must be simple to learn how to operate the game.	[56]

3.4. Postgame Questionnaire. As presented in Table 2, the postgame questionnaire assessed several dimensions of the cloud gaming experience, including appeal (AP), performance, intuitive controls (ICs), and learnability (LE). This stage aimed to gather feedback on game graphics, accessibility, game variety, and how these elements influenced user engagement. Additional questions were included to evaluate performance in terms of video quality, audio quality, and multimedia control. Participants rated each item using a 5-point Likert scale, where 1 = strongly disagree, 2 = disagree, 3 = undecided, 4 = agree, and 5 = strongly agree.

3.5. Posttest Questionnaire. As shown in Table 3, the posttest questionnaire focused on assessing input quality, controllability, responsiveness, immediate feedback, output quality, and overall player experience. The aim was to evaluate aspects such as audio clarity, video quality, game variety, device accessibility, and customer support. Participants rated each item on a Likert scale ranging from "strongly disagree" to "strongly agree." The methodology here reflects previous studies that have assessed cloud gaming QoE [39, 54].

All quality factors mentioned in the study were selected according to the recommendations of previous studies [39], ITU-T Recommendation G [2, 10, 46, 60, 65].

4. Results

The user survey, targeting gamers and industry experts, yielded over 62 responses. This feedback was analyzed to understand user needs and priorities regarding cloud gaming quality factors. While the sample size of 62 respondents is relatively small, it serves as a valuable exploratory dataset for this initial phase of research. The respondents represent a targeted group of experienced gamers, which allows for in-depth insight into user preferences and behaviors within this specific segment. Although the findings may not be broadly generalizable, they provide a solid foundation for future studies with larger and more diverse populations. This approach aligns with the findings of Shrivastava and Damle [8], who also highlighted the importance of a broad sample size to enhance the applicability of their results to the general gaming population.

4.1. Pretest Questionnaire. The pretest phase utilized a three-part questionnaire to identify expert participants. A broad initial pool was screened to assess gaming experience and determine eligibility. Subsequently, 42 expert participants (balanced in gender) were selected from a pool of 60 after excluding 18 beginners. The participants' birth years ranged from 1989 to 2008, reflecting the typical age range of gamers in recent studies [15]. This helps us understand the demographics of the respondents, as shown in Figure 2, depicting the gender of participants. The distribution of respondents' gaming experience levels in this study closely aligns with the findings from Basheer et al. [7], who also observed that the majority of active cloud gamers are intermediate or expert players.

In terms of gaming experience, Figure 3 presents the breakdown of participants' proficiency levels. The distribution of their experience level is significant, with 70% of participants

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Table 3: Posttest questionnaire.

Quality factor	Question	References
Input quality		
Controllability	• Users should be able to master and control game elements easily.	[57]
Responsiveness	• Users should experience a minimal delay between input commands and the corresponding actions in the game.	[58]
Immediate feedback	• Audio and visual effects must be in sync with real-time game progress.	[59]
Output quality		
Audio quality	• The sound must be clear, smooth, and without delay.	[31]
Video quality	• The video must be controllable, and its quality can be raised and lowered as needed.	[60]
Video discontinuity	• The video should not be interrupted, which affects the user experience.	[60]
Player experience		—
Library content	• The game library should have variety and focus on popular titles that cater to different preferences.	[61]
Accessibility	• Players should be able to access games via various devices. • An accurate description must be given of the minimum capabilities (hardware, network) that the player must have to obtain stable service.	[62]
Cost and pricing models	• There must be more than one pricing model available and transparent.	[46]
Social features	• Social features must be available, such as multiplayer capabilities and community forums.	[46]
Customer support	• Fast and effective customer support services must be available to help solve problems and provide solutions.	[63]
Security	• Data protection and privacy procedures must be ensured, especially in electronic payment operations.	[64]
Availability	• The system must be available 24/7.	[65]

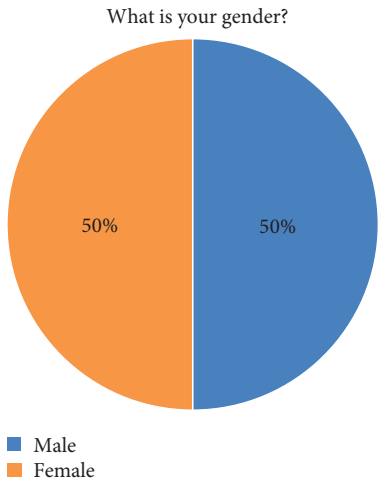

Figure 2: Gender of participants.

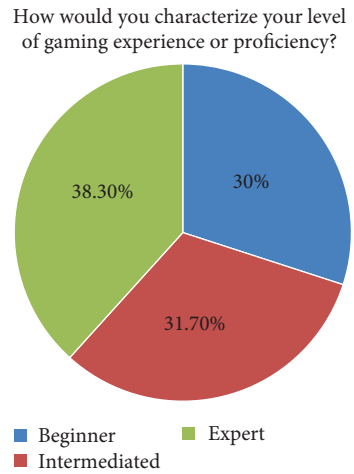

Figure 3: Player proficiency.

classifying themselves as either intermediate or expert. This finding is consistent with previous studies, such as those by Shrivastava and Damle [8], which report a similar concentration of intermediate and expert gamers within the cloud gaming community.

For weekly gaming hours, Figure 4 shows that 28.3% of participants play less than 5 h per week, 30% play 5–10 h, 25% play 10–20 h, and 16.7% are categorized as experts or "game addicts," dedicating more than 20 h per week. This result mirrors findings from previous research, such as the study by Moina-Rivera et al. [9], which found that more frequent gaming activity leads to stronger preferences for technical features like low latency and high-resolution graphics.

Additionally, Figure 5 breaks down the frequency of gaming sessions per week, showing that 50% of participants engage in daily or more frequent gameplay. This finding corroborates the research by Laghari et al. [31], who found that regular gaming sessions correlate with greater interest in performance-related features.

As for the type of devices used for gaming, Figure 6 shows that 25% of participants use a PC (desktop), while 60% use either a smartphone/tablet or a gaming console (PlayStation

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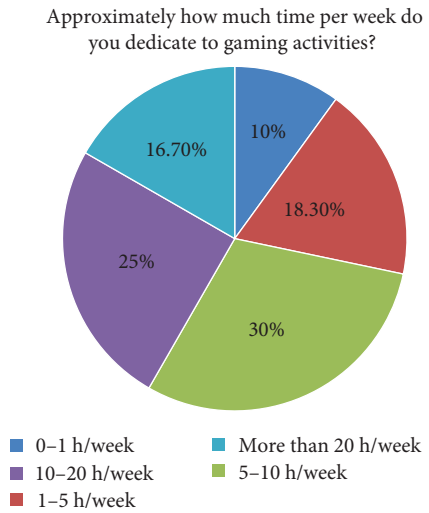

Figure 4: Player's experience/proficiency information.

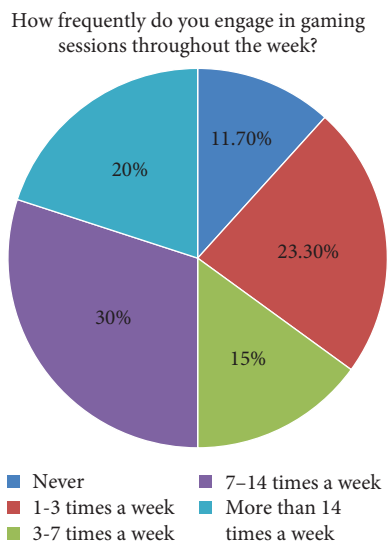

Figure 5: The number of sessions that the player joins each week.

annd Xbox). This device preference aligns with prior findings from Moina-Rivera et al. [9], who highlighted the growing dominance of smartphones and consoles in the gaming market due to their accessibility and affordability.

In the context of specific games tested in the study, Figures 7 and 8 present the experience levels of players for the games "RIDE 4" and "GTA". For "RIDE 4," 46.7% of players are classified as experts, while 43.3% are experts in "GTA." This is consistent with the general trends reported by Laghari et al. [15], who found that experienced players are often drawn to more complex and action-packed titles.

4.2. Postgame Questionnaire. The postgame questionnaire covers four main factors: performance indication (PI), LE, AP, and ICs. Each factor is assessed using a 5-point continuous scale, with 1 being "strongly disagree," 2 being "disagree," 3 being "neutral," 4 being "agree," and 5 being "strongly agree." These factors were partially adapted to the gaming domain,

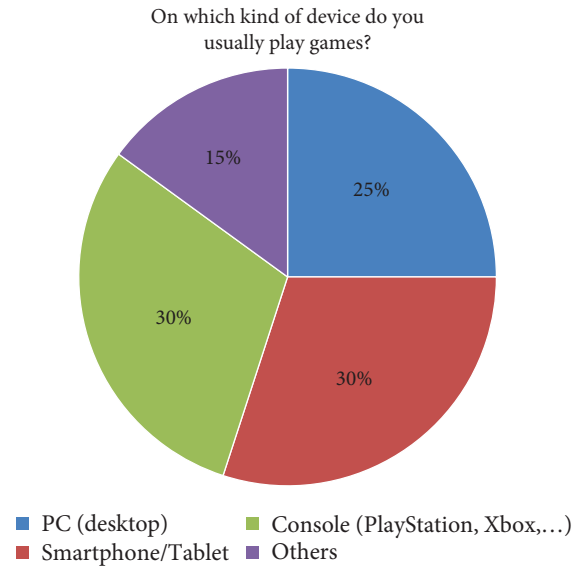

Figure 6: Type of devices used.

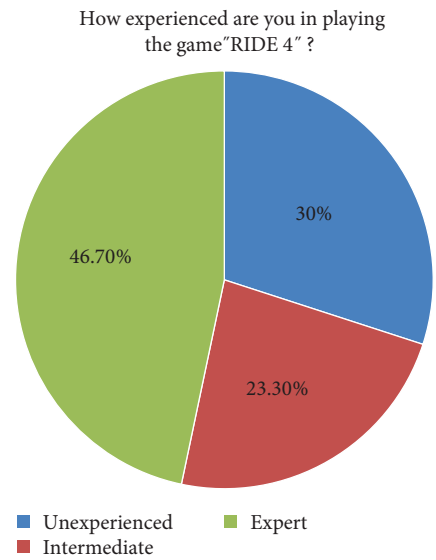

Figure 7: Mastering players for RIDE 4.

drawing from established scales like the Cognitive Absorption Scale (CAS) by Jumani et al. [57], the User Engagement Scale [66], and the PENS questionnaire.

4.2.1. AP. Figure 9 shows the first question under AP, where 42.9% of respondents strongly agreed that cloud gaming platforms should offer games with high-quality graphics. This was the highest percentage among the respondents.

Figure 10 reflects responses to the second AP question, with 45.2% strongly agreeing that cloud gaming platforms need to offer gaming access and smooth performance compared to PC gaming.

Figure 11 shows the third AP question, where 47.6% strongly agreed that cloud gaming platforms should offer a variety of game titles and subscription options to enhance their AP.

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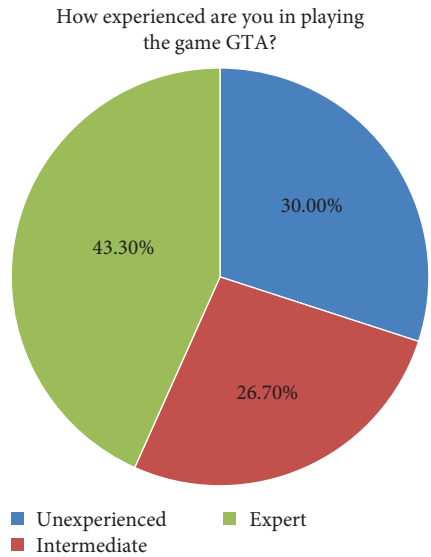

Figure 8: Mastering players for GTA.

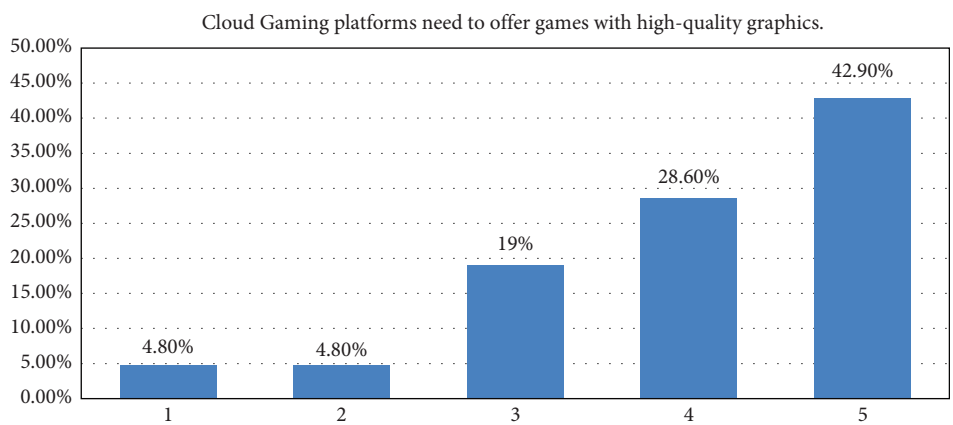

Figure 9: Appeal question-1.

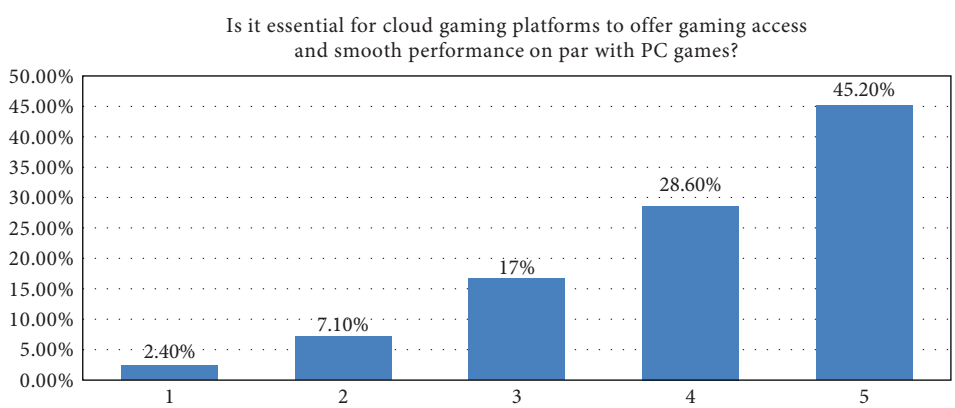

Figure 10: Appeal question-2.

The importance of high-quality graphics (AP) aligns with findings from previous research by Basheer et al. [7], where users identified graphics quality as one of the most critical factors when choosing cloud gaming services. This trend has remained consistent across several studies, indicating the strong consumer demand for visual fidelity in cloud gaming.

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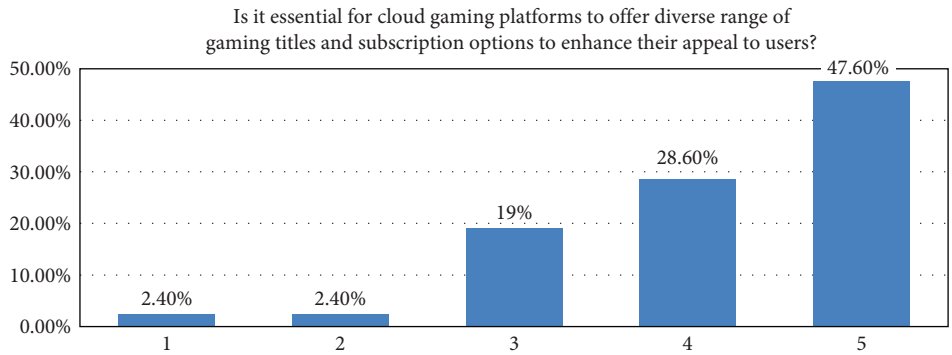

Figure 11: Appeal question-3.

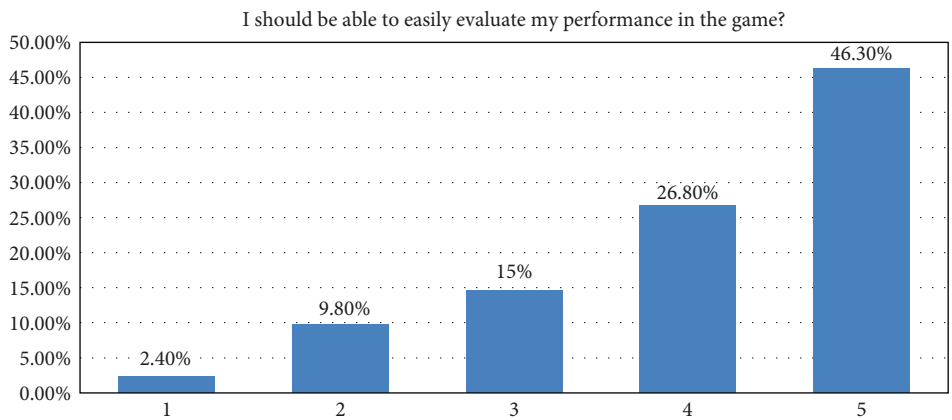

Figure 12: Performance indication question-1.

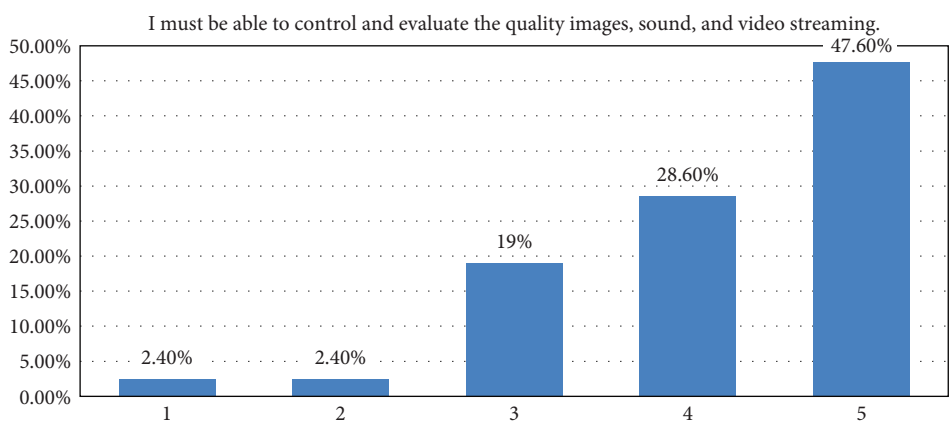

Figure 13: Performance indication question 2.

4.2.2. PI. Figure 12 presents results from the first PI question, where 46.3% strongly agreed that users should be able to easily evaluate their performance in the game.

Figure 13 reflects the second PI question, where 47.6% of respondents strongly agree that users should have the ability to control and assess the quality of images, sound, and video, emphasizing the importance of customization and optimization for individual preferences. Additionally, 28.6% express agreement with this sentiment, further highlighting the significance of empowering users with such control over their multimedia experience. Meanwhile, 19% of participants remain neutral, suggesting a potential need for clarification or further exploration of the benefits and implications of this functionality.

Figure 14 displays responses to the third performance question, where 47.6% strongly agreed that cloud gaming should perform consistently overtime, avoiding issues like stuttering or disconnections.

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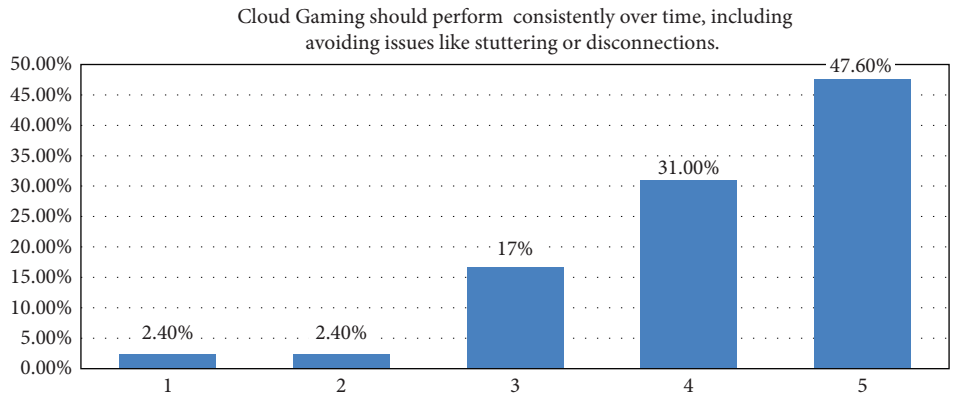

Figure 14: Performance indication question-3.

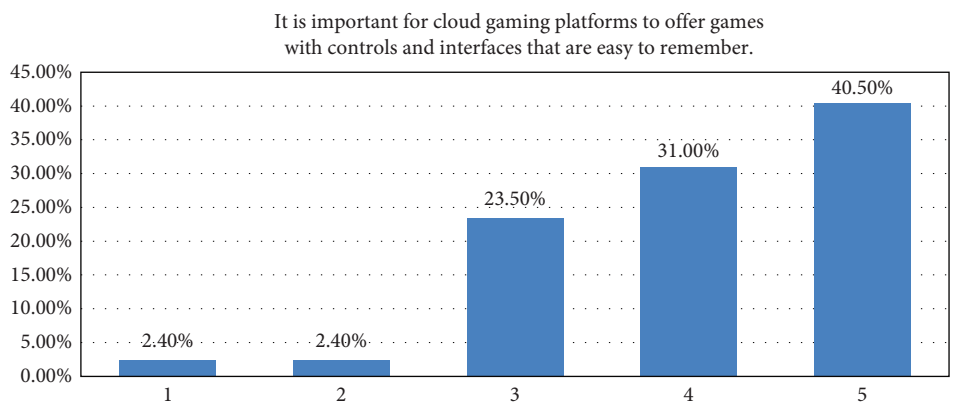

Figure 15: Intuitive controls question-1.

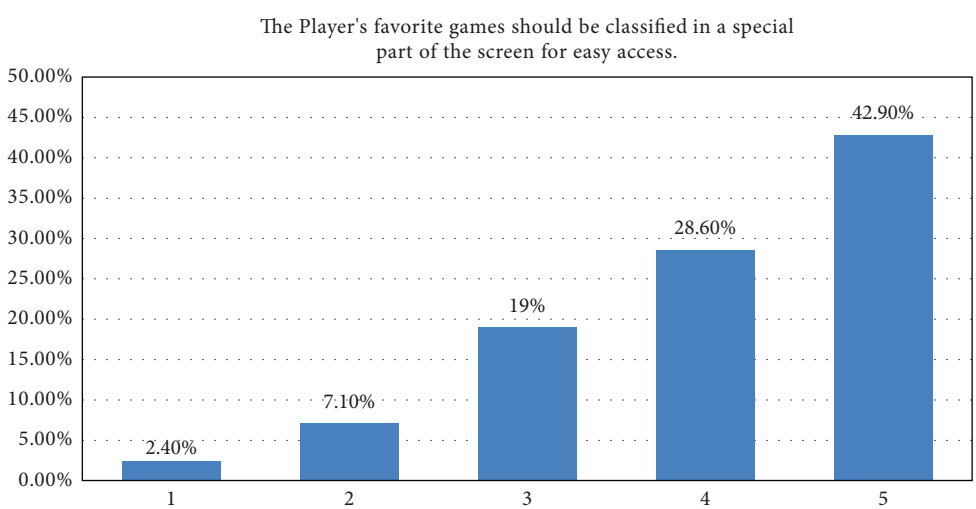

Figure 16: Intuitive controls question-2.

These results are consistent with previous studies by Shrivastava and Damle [8] and Laghari et al. [31], who emphasized the importance of performance indicators and user control over video and audio settings in improving overall user experience in cloud gaming environments.

4.2.3. ICs. Figure 15 shows that 40.5% strongly agreed that cloud gaming platforms should offer games with easy-to-remember controls and intefaces.

Figure 16 shows 42.9% of respondents strongly agreed with classifying the player's favorite games in a dedicated section of

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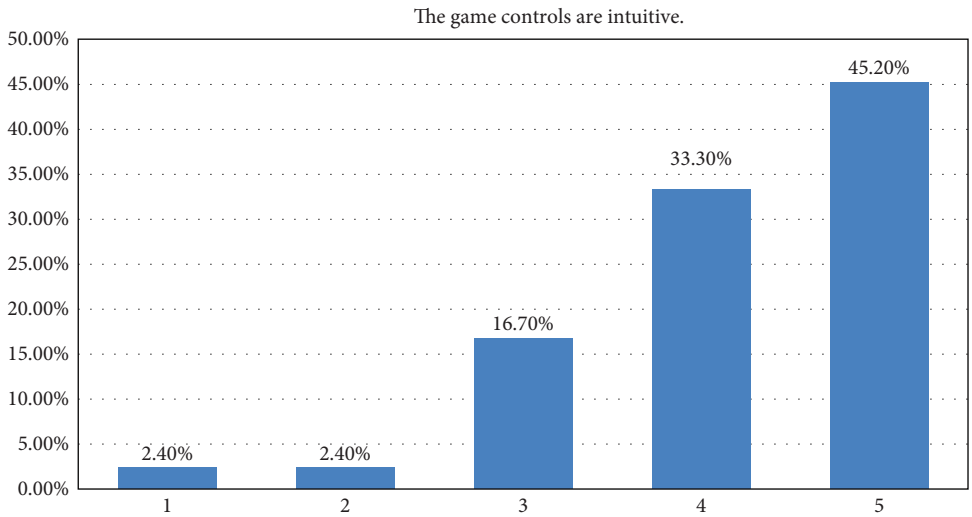

Figure 17: Intuitive controls question-3.

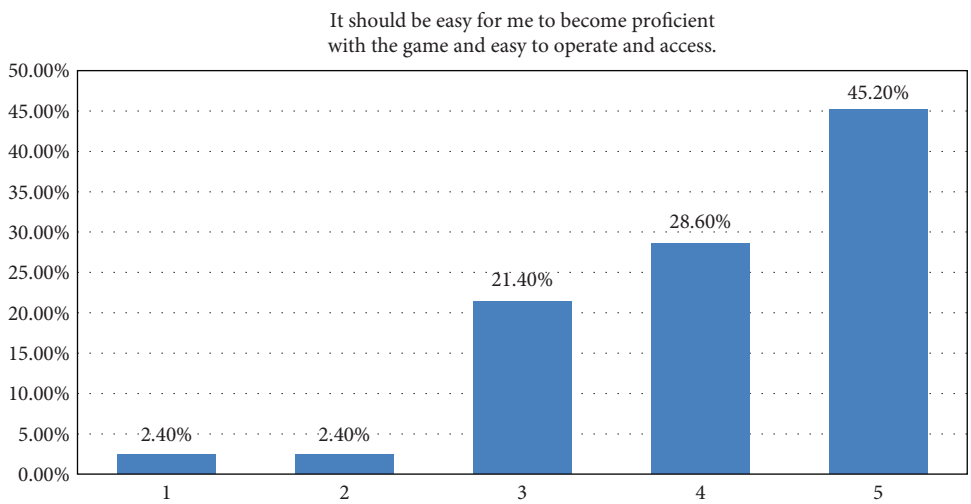

Figure 18: Learnability question-1.

the screen, emphasizing the convenience of quick and easy access to preferred titles. Additionally, 28.6% agree with this proposal, further supporting prioritizing accessibility for favored games. Meanwhile, 19% of participants remain neutral, indicating a potential need for further exploration or clarification on the benefits of this feature.

Figure 17 highlights that 45.2% of respondents strongly agree that intuitive game controls are essential, emphasizing their critical role in enhancing the gaming experience. Another 33.3% agree with this sentiment, reinforcing the importance of user-friendly interfaces. Meanwhile, 16.7% of participants remain neutral, suggesting a potential opportunity for improvement or clarification in understanding the significance of ICs.

This preference for ICs is consistent with findings from Jumani et al. [57], who demonstrated that user-friendly interfaces are crucial to player satisfaction and engagement. Additionally, the study by Laghari et al. [46] found that accessibility and ease of navigation are highly valued in cloud gaming services, further confirming the findings of this study.

4.2.4. LE. Figure 18 displays that 45.2% strongly agree that it should be easy for users to become proficient with the game and easy to operate and access, and 28.6% agree, 21.4% of the respondents are neutral.

Figure 19 highlights that 47.8% strongly agree that comments and prompts for all games in the library should be provided to users to help them learn game mechanics, objectives, and strategies. And 33.3% agree and 16.7% of the respondents are neutral.

Figure 20 shows that 47.6% strongly agree that it must be simple to learn how to operate the game, 31% agree, and 19% of the respondents are neutral.

The findings in this study support previous research by Möller et al. [56], who stressed the importance of easy LE and accessibility in cloud gaming platforms to ensure a positive user experience, particularly for new gamers.

Table 4 summarizes the results of the postgame questionnaire, showing the percentage of responses for each question and factor.

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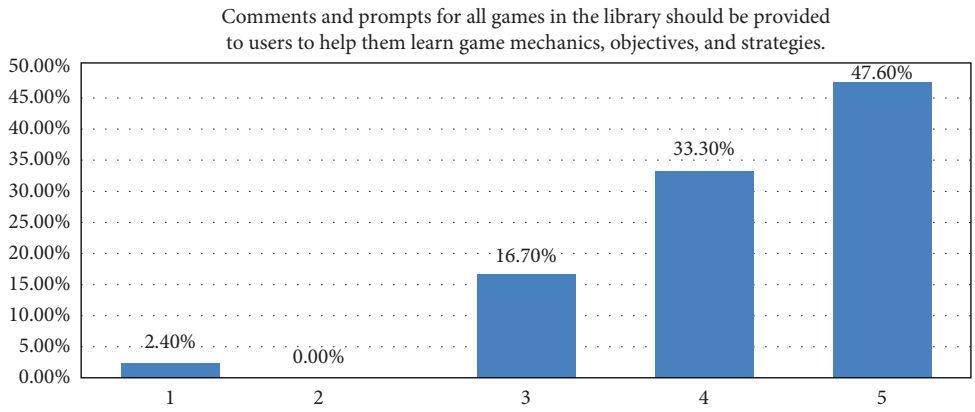

Figure 19: Learnability question-2.

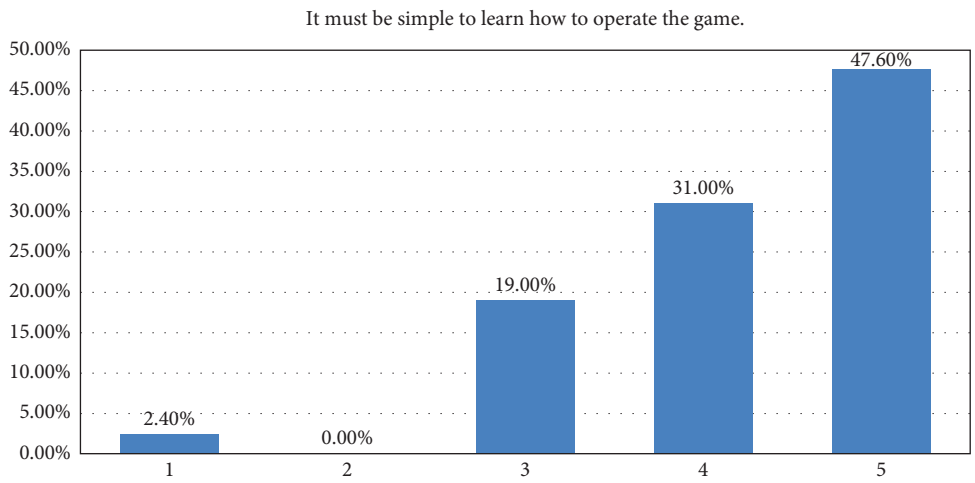

Figure 20: Learnability question-3.

Table 4: Postgame questionnaire result summary.

Quality factor	Question	Agree (%)	Strongly agree (%)	Total (%)
1. Appeal		**28.2**	**45.2**	**73.4**
	• How important are cloud gaming platforms to offer games with high-quality graphics?	28.6	42.9	71.5
	• It is essential that cloud gaming platforms provide gaming accessibility and smooth performance on par with PC gaming.	28.6	45.2	73.8
	• It is imperative that cloud gaming platforms offer a variety of game titles and subscription options to enhance their appeal to users.	27.6	47.6	75.2
2. Performance indication		**28.8**	**47.1**	**75.9**
	• I should be able to evaluate my performance in the game easily.	26.8	46.3	73.1
	• I must be able to control and evaluate the quality of images, sound, and video streaming.	28.6	47.6	76.2
	• Cloud gaming should perform consistently over time, including avoiding issues like stuttering or disconnections.	31	47.6	78.6
3. Intuitive controls		**30.9**	**42.8**	**73.7**
	• It is important for cloud gaming platforms to offer games with controls and interfaces that are easy to remember.	31	40.5	71.5
	• The player's favorite games should be classified in a special part of the screen for easy access.	28.6	42.9	71.5
	• The game controls are intuitive.	33.3	45.2	78.5

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Table 4: Continued.

Quality factor	Question	Agree (%)	Strongly agree (%)	Total (%)
4. Learnability		**30.1**	**47.6**	**77.7**
	• It should be easy for me to become proficient with the game and easy to operate and access.	28.5	45.2	73.7
	• Comments and prompts for all games in the library are provided to users to help them learn game mechanics, objectives, and strategies.	33.5	47.6	81.1
	• It must be simple to learn how to operate the game.	28.7	50	78.7

Note: Bold values indicate the average for each factor.

Table 5: ANOVA results for demographic differences in quality preferences.

Factor	*F*-value	*p*-Value	Significance
Appeal (high-quality graphics)	5.43	0.008	Significant ($p<0.01$)
Performance indication	3.25	0.023	Significant ($p<0.05$)
Intuitive controls	2.90	0.051	Marginally Significant ($p=0.051$)
Learnability	4.12	0.014	Significant ($p<0.05$)

Table 6: Regression analysis for predicting user satisfaction.

Variable	Beta	Standard error	*t*-Value	*p*-Value
Gaming experience (expert)	0.25	0.12	2.08	0.040
Device type (console)	0.18	0.09	2.00	0.049
Weekly gaming hours	0.30	0.08	3.75	<0.001
Adjusted R^2	0.56	—	—	—

4.3. Inferential Statistics Analysis (ANOVA and Regression). To explore demographic differences in user preferences, inferential statistical methods such as ANOVA and regression were applied. These tests were used to analyze how gaming experience, device type, and weekly gaming hours influenced preferences for cloud gaming quality factors including AP, PI, LE, and ICs [67].

4.3.1. ANOVA Results. An ANOVA was conducted to examine the impact of gaming experience (beginner, intermediate, and expert) and device type (PC, console, and mobile) on key quality factors [68]. The results, as shown in Table 5, revealed significant differences in preferences between experience levels and device types for certain quality factors.

- AP: Significant differences were found between expert players and beginners. Expert players were more likely to rate high-quality graphics as a critical factor.
- PI: Users who engaged in more than 20 h of gaming per week expressed a greater need for features that enable easy performance evaluation.
- ICs: Marginally significant differences were observed. Casual gamers tended to value ICs more, likely due to less experience with complex input schemes.
- LE: Significant differences were found. Novice users emphasized LE, highlighting the importance of clear tutorials and easy onboarding for new players.

4.3.2. Regression Analysis. A regression analysis was conducted to identify predictors of user satisfaction in cloud gaming. The analysis, as presented in Table 6, included variables like gaming experience, device type, and weekly gaming hours to predict overall satisfaction.

The regression analysis revealed that gaming experience, device type, and weekly gaming hours significantly predicted overall user satisfaction with cloud gaming platforms (adjusted $R^2=0.56$). Specifically, expert gamers were significantly more likely to report higher satisfaction ($\beta=0.25$, $p=0.040$) and console users showed similarly positive associations ($\beta=0.18$, $p=0.049$). Weekly gaming hours had the strongest effect ($\beta=0.30$, $p<0.001$), suggesting that more frequent gamers tend to be more satisfied with their cloud gaming experience. While these results are promising, further analysis is needed to explore how these variables specifically influence subfactors such as LE or PI. Future studies should incorporate larger and more diverse samples and qualitative methods like interviews to validate and deepen these findings.

4.4. Posttest Questionnaire. In the posttest phase, participants were asked to evaluate quality elements based on their experience with the cloud gaming service, previous gaming experience, and expectations as users of the service. They rated these elements on a 5-point scale, from "strongly disagree" to "strongly agree."

Table 7 outlines the main and subfactors included in the posttest questionnaire, including input quality (controllability,

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Table 7: Main and subfactors of postgame questionnaire.

Main factors	Subfactors
Input quality	Controllability, responsiveness, immediate feedback
Output quality	Audio quality, video quality, video discontinuity
Player experience	Library content, accessibility, cost and pricing models, social features, customer support
Other factors	Security, availability

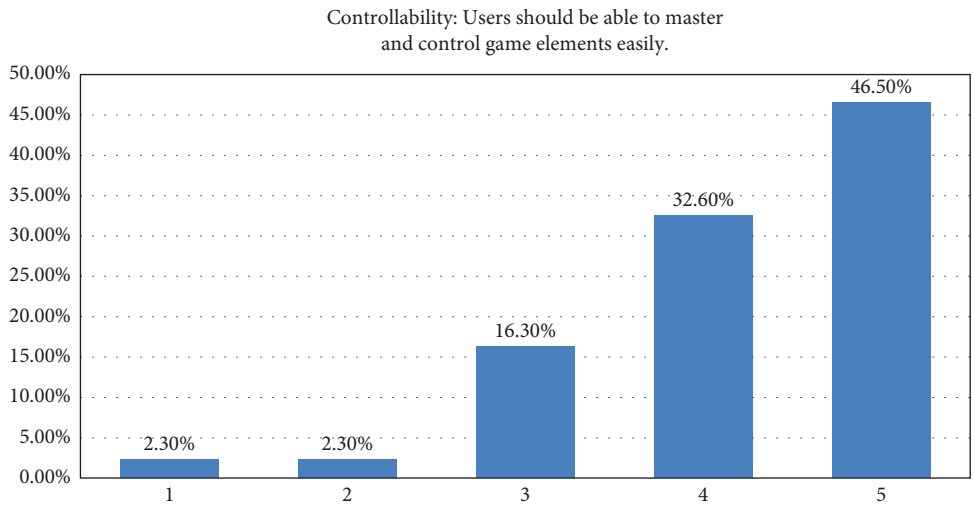

Figure 21: Controllability question result.

responsiveness, and immediate feedback), output quality (audio quality, video quality, and video discontinuity), and player experience (library content, accessibility, cost and pricing models, social features, and customer support).

The component scores of each aspect are computed as the average value of its items.

4.4.1. Input Quality. Figure 21 shows that 46.5% strongly agree that to achieve controllability in cloud gaming, users should be able to master and control game elements easily. 32.6% agree and 16.3% of the respondents are neutral.

Figure 22 displays that 45.2% strongly agree that to achieve responsiveness, users should experience a minimal delay between input commands and the corresponding actions in the game, 31% agree, and 19% of the respondents are neutral.

Figure 23 shows that 46.5% strongly agree that audio and visual effects must be in sync with real-time game progress, 34.9% agree, and 14% of the respondents are neutral.

4.4.2. Output Quality. Figure 24 shows that 48.8% strongly agree that the sound must be clear, smooth, and without delay, 27.9% agree, and 18.6% of the respondents are neutral.

Figure 25 highlights that 46.5% strongly agree that video streaming must be controllable and its quality can be raised and lowered as needed, 34.9% agree, and 14% of the respondents are neutral.

Figure 26 shows that 46.5% strongly agree that the video should not be interrupted throughout the game session, 32.6% agree, and 16.3% of the respondents are neutral.

4.4.3. Player Experience. Figure 27 shows that 48.8% strongly agree that the game library should have variety and focus on popular titles that cater to different preferences, 27.9% agree, and 18.6% of the respondents are neutral.

Figure 28 shows that 48.8% strongly agree players should be able to access games via various devices, 30.2% agree, and 16.3% of the respondents are neutral.

Figure 29 presents that 41.9% strongly agree an accurate description must be given the minimum capabilities (hardware and network) that the player must have to obtain stable service, 37.2% agree, and 16.3% of the respondents are neutral.

Figure 30 shows that 44.2% strongly agree that cost and pricing must be more than one pricing model available and transparent, 34.9% agree, and 16.3% of the respondents are neutral.

Figure 31 shows that 53.5% strongly agree that social features must be available, such as multiplayer capabilities and community forums, 30.2% agree, and 11.3% of the respondents are neutral.

Figure 32 highlights that 46.5% strongly agree that fast and effective customer support services must be available to help solve problems and provide solutions, 30.2% agree, and 18.6% of the respondents are neutral.

4.4.4. Other Factors. Figure 33 shows that 53.5% strongly agree that cloud gaming providers must ensure data protection and privacy procedures, especially in electronic payment operations, 25.6% agree, and 16.3% of the respondents are neutral.

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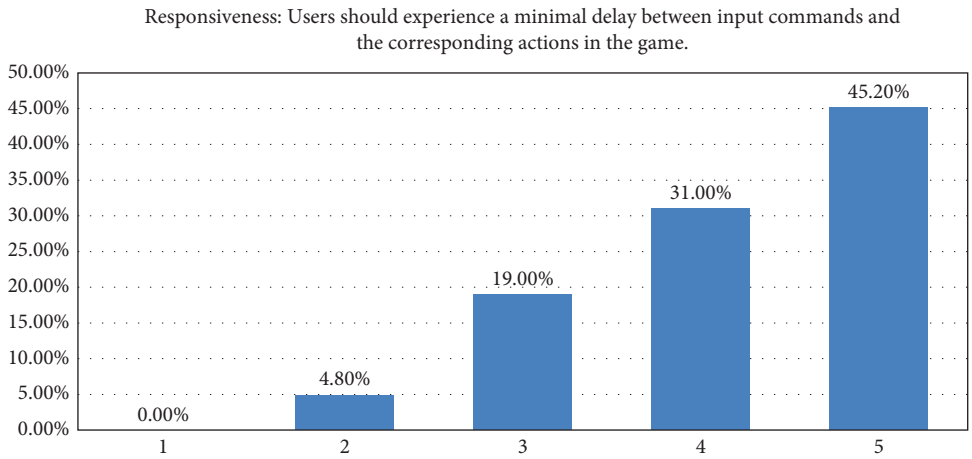

Figure 22: Responsiveness question result.

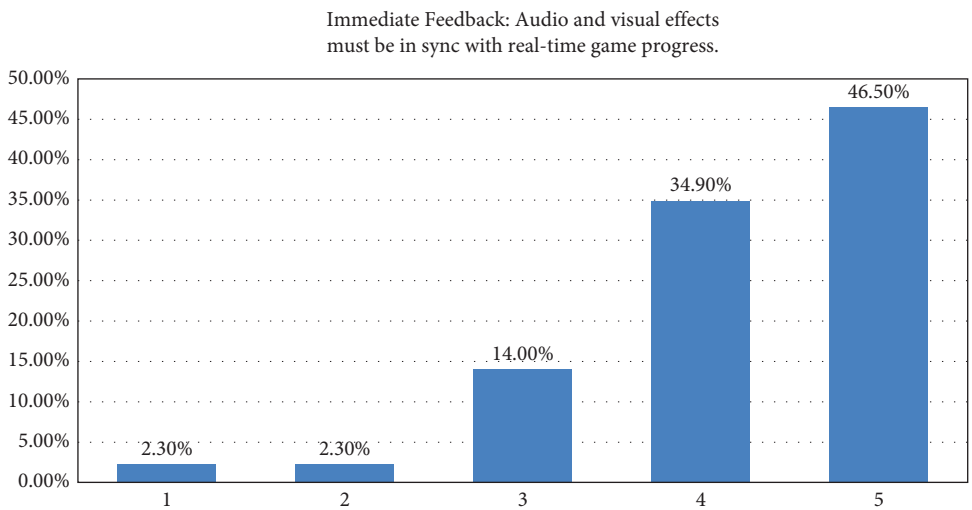

Figure 23: Immediate feedback question result.

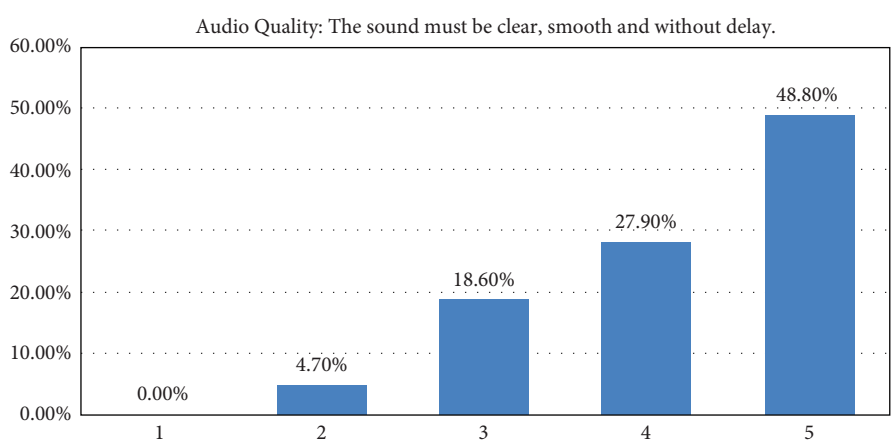

Figure 24: Audio quality question result.

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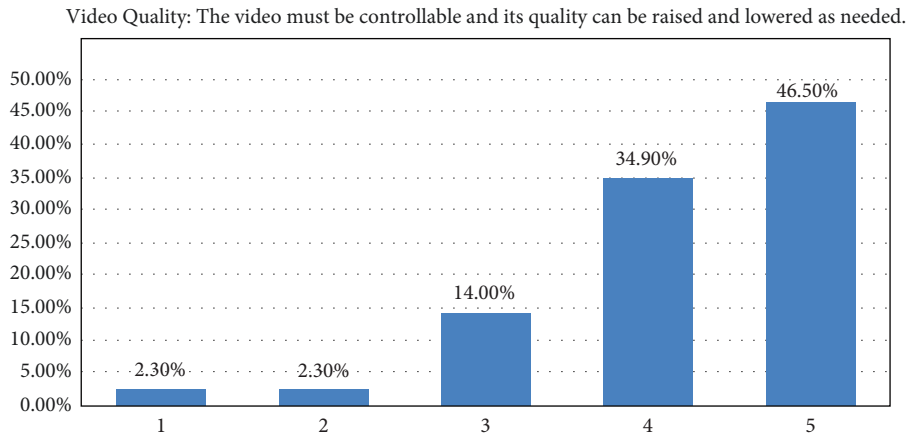

Figure 25: Video quality question result.

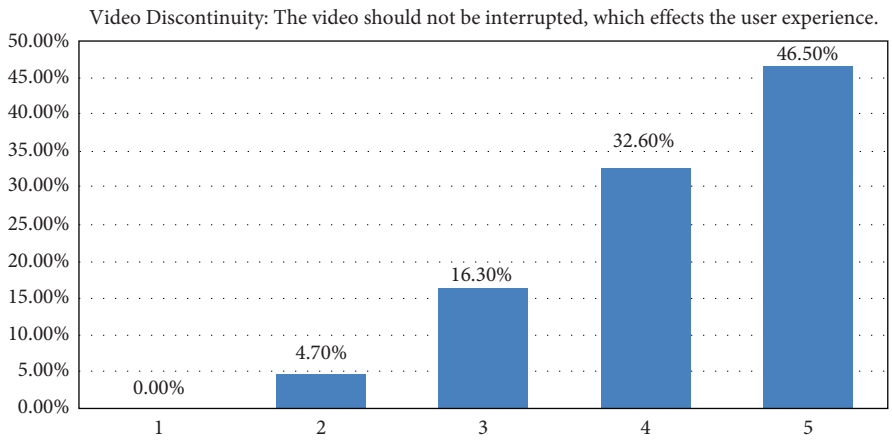

Figure 26: Video discontinuity question result.

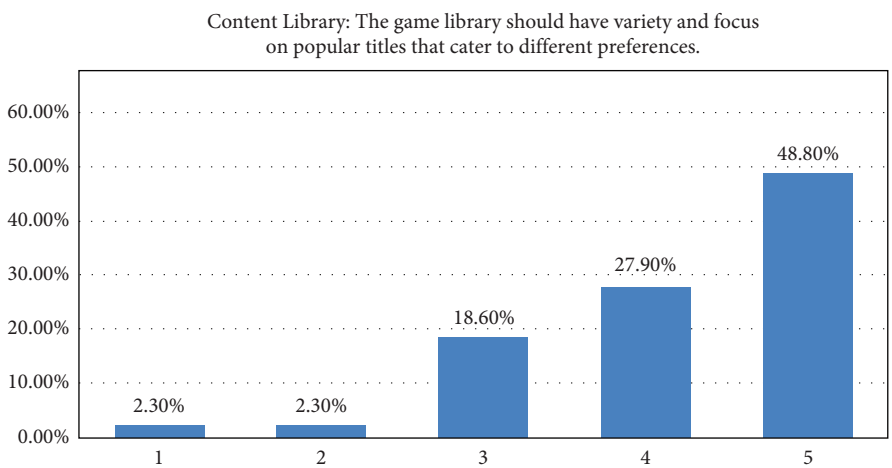

Figure 27: Content library question result.

Figure 34 shows that 50% strongly agree that the system must be available 24/7, 28.6% agree, and 16.7% of the respondents are neutral.

Table 8 summarizes the posttest questionnaire results, showing the percentages of agreement for each factor.

The percentage that did not support the study was coordinated to conduct future interviews with descriptive questions to express their own opinions.

While this study focused on capturing user perceptions of GaaS quality, it did not include objective technical validation

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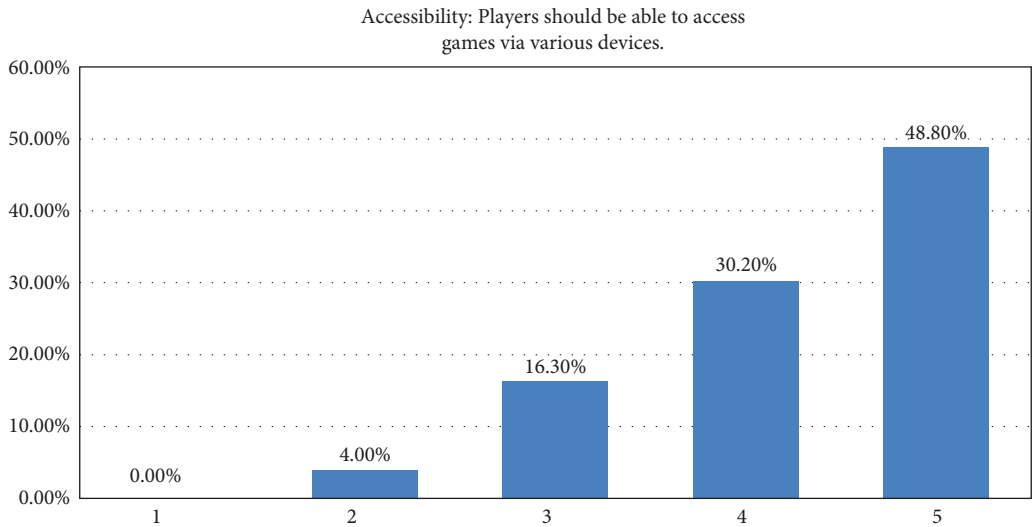

Figure 28: Accessibility question-1 result.

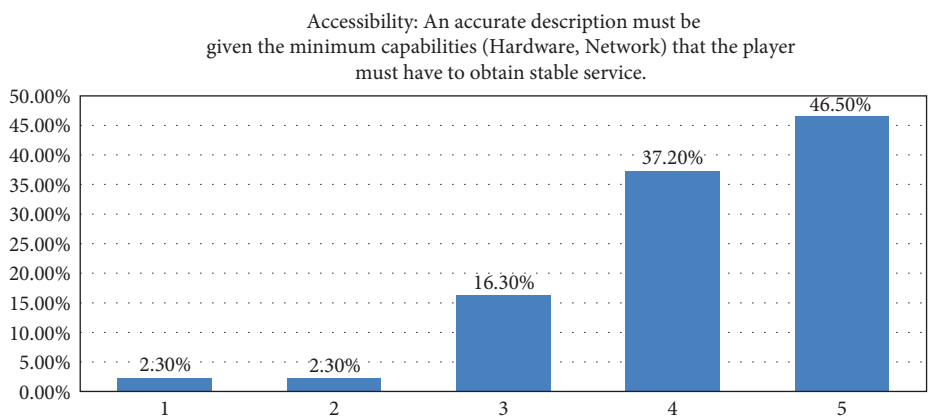

Figure 29: Accessibility question-2 result.

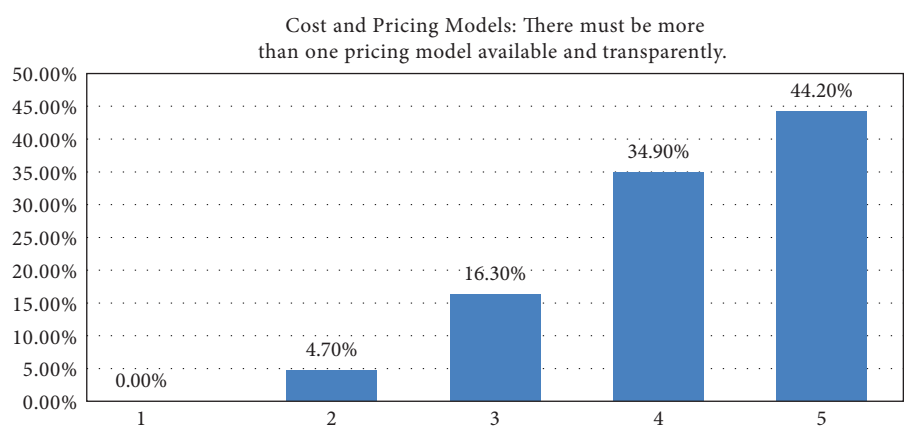

Figure 30: Cost and pricing question result.

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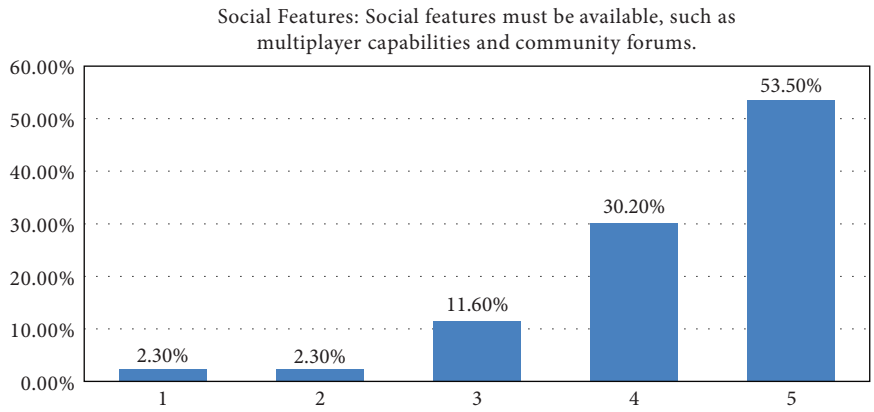

Figure 31: Social features question result.

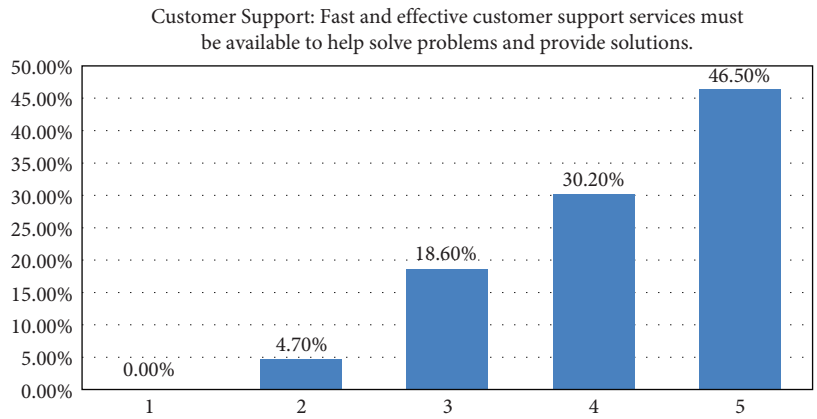

Figure 32: Customer support question result.

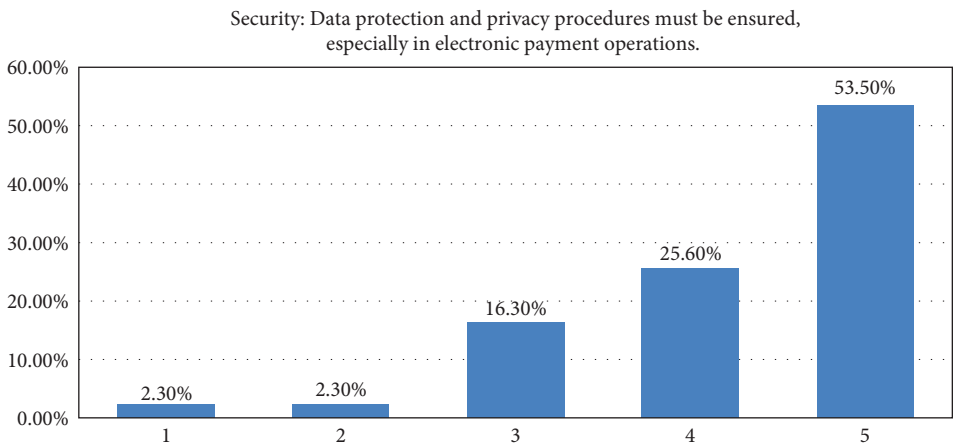

Figure 33: Security question result.

metrics such as latency, jitter, or frame rate stability. These factors, while indirectly captured through questions related to responsiveness and performance, remain essential components of QoE. Future research should combine user-centric data with system-level measurements using tools such as network profilers or cloud monitoring utilities to validate and triangulate findings more comprehensively.

5. Discussion

This study employed a survey-based methodology to assess user preferences and evaluate the performance of cloud gaming platforms, specifically focusing on factors that contribute to user satisfaction. The results gathered from 62 respondents revealed several key insights into what gamers prioritize

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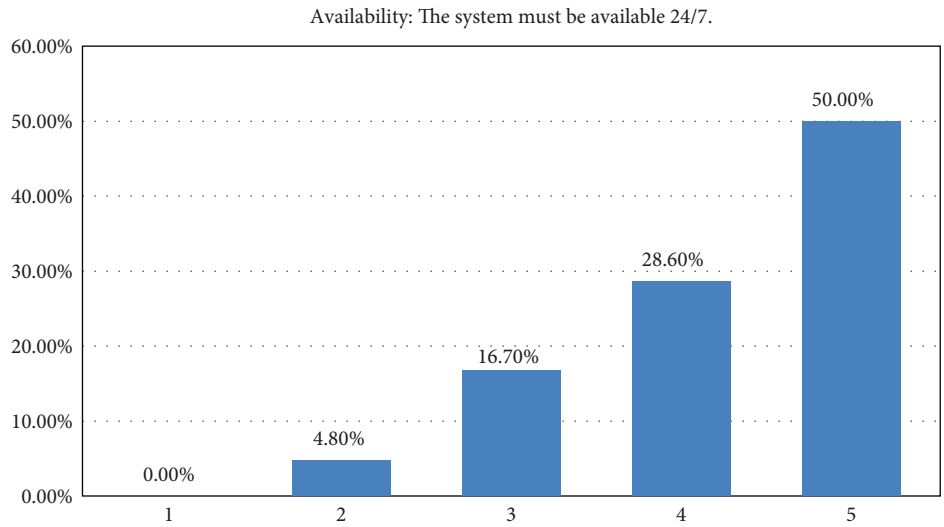

Figure 34: Availability question result.

Table 8: Posttest questionnaire result summary.

Quality factor	Question	Agree (%)	Strongly agree (%)	Total (%)
Controllability	Users should be able to master and control game elements easily.	32.6	46.5	79.1
Responsiveness	Users should experience a minimal delay between input commands and the corresponding actions in the game.	31	45.2	76.2
Immediate feedback	Audio and visual effects must be in sync with real-time game progress.	34.9	46.6	81.5
Audio quality	The sound must be clear, smooth, and without delay.	27.9	48.8	76.7
Video quality	The video must be controllable and its quality can be raised and lowered as needed.	34.6	46.7	81.3
Video discontinuity	The video should not be interrupted, which affects the user experience.	32.6	46.5	79.1
Content library	The game library should have variety and focus on popular titles that cater to different preferences.	27.9	48.8	76.7
Accessibility	Players should be able to access games via various devices.	30.2	48.9	79.1
	An accurate description must be given the minimum capabilities (hardware, network) that the player must have to obtain stable service.	37.2	41.9	79.4
Cost and pricing models	There must be more than one pricing model available and transparently.	34.9	44.2	79.1
Social features	Social features must be available, such as multiplayer capabilities and community forums.	30.2	52.5	82.7
Customer support	Fast and effective customer support services must be available to help solve problems and provide solutions.	30.3	46.5	76.8
Security	Data protection and privacy procedures must be ensured, especially in electronic payment operations.	25.6	53.6	79.2
Availability	The system must be available 24/7.	28.6	50	78.6

when using GaaS platforms. Over 70% of respondents expressed satisfaction with aspects such as high-quality graphics, game library diversity, and smooth performance. Additionally, users highly valued features that enabled them to monitor their gaming performance and adjust audio/video settings, ensuring a consistent gaming experience. The importance of user-friendly interfaces, ICs, and accessible learning resources was also highlighted, along with the growing need for social features, such as multiplayer options and community forums, to foster engagement.

One of the most significant findings was the critical role of security. Data protection and user privacy especially during financial transactions emerged as top concerns for gamers. These concerns reinforce the need for robust security protocols to build

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and maintain trust in cloud gaming platforms. As data breaches and cybersecurity threats continue to rise, ensuring a secure environment for users is crucial for the long-term success of GaaS.

Despite these positive aspects, the results also pointed to a notable gap between user needs and the current capabilities of GaaS platforms. Users expressed concerns about the clarity of communication regarding minimum system requirements and network conditions. This lack of transparency can hinder the user experience, especially when users encounter performance issues due to unoptimized hardware or network connections. Addressing these concerns is crucial for platforms looking to meet user expectations effectively. Future platforms should invest in clear documentation and intuitive onboarding processes that guide users through system requirements and setup.

Our findings suggest that cloud gaming platforms should focus on several areas for improvement. Transparency is a key issue; platforms need to provide clear and accessible information about minimum system requirements, pricing structures, and network conditions. Improving social features, such as in-game chat, multiplayer options, and community forums, will enhance user engagement. Additionally, ensuring accessibility across various devices, including PCs, smartphones, and gaming consoles, will help platforms cater to a broader audience. Data protection and privacy measures must also be continuously improved to maintain user trust, particularly as concerns around cybersecurity in online platforms grow.

Despite the importance of technical performance, platforms must also consider user-centric design features, such as ICs and learning resources. By enabling users to better understand game mechanics and platform functionality, platforms can further optimize the learning experience. Performance consistency, including minimizing latency, packet loss, and ensuring stable frame rates, is also critical to creating a seamless gaming experience. Clear communication about these technical requirements can significantly enhance user satisfaction and reduce frustrations linked to inconsistent performance.

Beyond addressing current challenges, emerging technologies hold significant promise for enhancing user satisfaction in cloud gaming. AI-powered systems can provide tailored gaming experiences by analyzing user preferences, behavior, and gaming habits. This technology can adjust game features in real-time, such as difficulty levels, recommended titles, and even control schemes based on individual preferences. AI can also enhance gameplay suggestions, offering players new games they might enjoy, based on their previous choices, similar to recommendation systems seen in streaming platforms like Netflix and Spotify. By leveraging AI, platforms can ensure that the gaming experience evolves alongside the player, providing increasingly personalized and immersive interactions.

Another major technological advancement is adaptive streaming, which adjusts video quality and resolution based on a user's current internet speed and network conditions. This technology allows cloud gaming platforms to maintain performance by dynamically optimizing bit rate and frame rates, ensuring that players with slower or fluctuating internet connections can still enjoy smooth gameplay. Adaptive streaming addresses a common pain point in cloud gaming, where performance issues such as lag and stuttering can occur when network conditions are not ideal. By incorporating adaptive streaming technology, platforms can significantly improve QoE and ensure a more consistent performance across varied network environments.

The quality model proposed here seems to strike a balance between actual technical service delivery and user expectations of quality. It captures subjective measures of satisfaction and translates these into design and performance targets for action through easily measurable performance dimensions. User reports about responsiveness can be easily translated into measurable technical parameters, such as input lag or latency in a game. These two aspects can be quantified and assessed by development teams. Subjective attributes such as the AP and accessibility of content can serve as valuable guides for content curation and UI/UX design. This connection enables GaaS providers to align the enhancement of infrastructure, software features, and performance-related service level agreements with user preferences.

In a theoretical context, this study has implications for human–computer interaction (HCI) and user experience (UX) design by defining the factors that influence QoE specifically within the realm of GaaS. The text combines psychological and behavioral constructs, such as immersion, control, and satisfaction, into key performance indicators (KPIs) to align experience-centered designs with technical implementations. This user-centric model can serve as a foundational point for future research on the quality of interactive systems and the delivery of adaptive services.

While the model demonstrates internal consistency based on survey findings, it has not yet been externally validated across diverse populations or deployment contexts. Additionally, qualitative methods such as user interviews or focus groups could provide richer insights and help refine the model's applicability. Future research should incorporate cross-context validation and triangulation to enhance generalizability and robustness.

However, despite the promise of these emerging technologies, several challenges must be addressed to fully realize their potential. One of the major barriers is the requirement for substantial data collection. AI-driven personalization relies on access to user data, which raises concerns about privacy and data security. As cloud gaming platforms gather data to fine-tune recommendations and experiences, it will be essential to balance personalization with data protection to maintain user trust.

Moreover, the success of adaptive streaming is contingent upon the network infrastructure available to users. In regions with low bandwidth or unstable connections, even adaptive streaming may not be sufficient to guarantee smooth gameplay. Research into next-generation networks, including 5G and edge computing, may hold the key to overcoming these challenges and providing a more stable and consistent experience for players worldwide.

Longitudinal studies are also needed to examine how user preferences evolve. Tracking changes in satisfaction and engagement with GaaS platforms can help developers anticipate future needs and trends. Additionally, future research should explore cross-cultural differences in gaming preferences, as varying cultural contexts may shape user experiences and expectations differently.

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In summary, this study highlights the critical quality factors that influence user satisfaction with cloud gaming platforms. The findings emphasize the importance of performance consistency, user-friendly design, and social features, while also underscoring the need for transparent communication and robust security measures. The incorporation of emerging technologies, such as AI-driven personalization and adaptive streaming, can significantly enhance the gaming experience, addressing both performance-related issues and individual user preferences. By leveraging these technologies, GaaS providers can enhance their offerings, close the gap between technical capabilities and user needs, and deliver a more user-centric experience. The future of GaaS lies in optimizing both the technical aspects and the user experience to provide seamless, engaging, and secure gaming environments.

6. Conclusion

This study significantly advances understanding of user satisfaction in GaaS by developing a comprehensive and user-centric quality model tailored to cloud gaming platforms. The model highlights critical quality factors such as controllability, accessibility, performance stability, social features, and data security that directly influence QoE and engagement. Survey results confirmed users prioritize high-resolution graphics, diverse game libraries, responsive controls, and seamless audio–visual synchronization, aligning well with the model. The study also underscores the need for greater service transparency. Users expressed concerns over unclear hardware requirements, network specifications, and pricing, which can cause frustration, especially for less tech-savvy or new users. Clear documentation and accessible onboarding are essential to address these issues and ensure consistent experiences. Security and privacy emerged as paramount, with users expecting robust data protection and secure payment systems to build trust and ensure platform viability. Emerging technologies offer promising enhancements: AI-driven personalization can tailor gaming experiences by recommending games and optimizing controls, while adaptive streaming adjusts bit rate and resolution in real time to reduce latency. However, these advances pose challenges around data collection, consent, and infrastructure that require careful governance. The proposed model bridges the gap between user expectations and current cloud gaming capabilities by combining objective metrics with subjective perceptions, providing a practical framework for developers. Though insightful, the study's sample size and scope suggest future research should expand samples, explore longitudinal trends, and consider cross-cultural differences to refine the model further. In sum, this model equips GaaS providers to enhance platform performance, trust, and user satisfaction, while thoughtful integration of new technologies promises a more personalized and robust gaming experience.

Appendix A

Appendix A1: Player Experience Models. The game experience questionnaire (GEQ) has been a key tool in gaming research, providing insights into various aspects of the gaming experience. Developed by Suznjevic et al. [49], the GEQ is structured to capture subjective experiences like immersion, enjoyment, flow, tension, and challenge. Numerous studies have harnessed the power of the GEQ to delve into player experiences across diverse game genres, including casual, action, strategy, and role-playing games [69]. Researchers have utilized the GEQ to probe the impact of game design elements, narrative structures, audio–visual presentation, and interaction mechanics on player engagement and enjoyment.

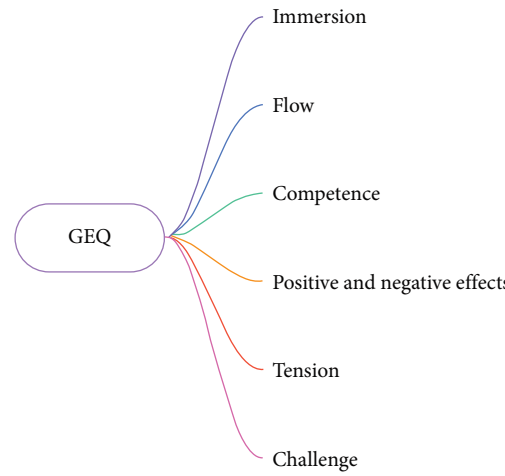

Figure A1: Game experience questionnaire.

Adaptations and validations of the GEQ for specific contexts or cultural groups emphasize its versatility and applicability across diverse gaming populations [69]. Additionally, researchers have employed the GEQ to assess interventions aimed at enhancing the gaming experience, such as the integration of novel game mechanics, virtual reality (VR) systems, or user interface designs. This flexibility underscores the utility of the GEQ in understanding the diverse factors that shape the gaming experience across a wide range of platforms and genres. Figure A1 illustrates the main Quality Aspects of GEQ as outlined by Tao et al. [70].

Appendix A2: Player Experience and Need Satisfaction (PENS) Questionnaire. The PENS Questionnaire is a tool used in psychology and game studies to evaluate various aspects of player experience and psychological needs fulfillment within gaming contexts [71]. Developed by Ryan et al. [55], the PENS questionnaire is designed to assess player experiences in terms of autonomy, competence, relatedness, immersion, and enjoyment.

The questionnaire includes items specifically designed to capture players' feelings of autonomy, or the sense of volition and choice within the gaming environment. It also assesses competence, referring to the player's perceived ability to effectively navigate and succeed within the game. Relatedness, another critical dimension, explores the player's sense of connection and social interaction within the gaming community or the game world [72]. Immersion refers to the depth of involvement and engagement experienced by the player, while enjoyment reflects the overall positive effect associated with gameplay [73].

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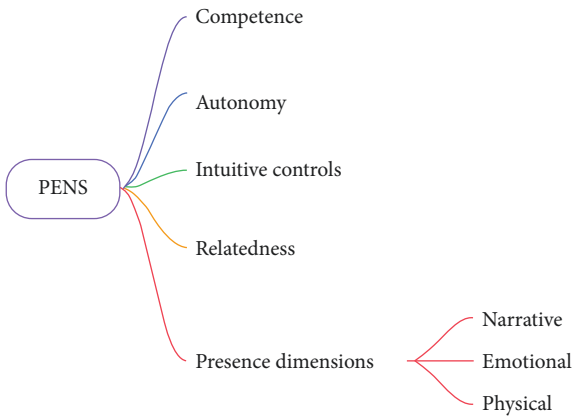

Figure A2: Player Experience and Need Satisfaction (PENS) Questionnaire.

Researchers and practitioners use the PENS Questionnaire to investigate how various game design features, mechanics, and social elements influence player experiences and psychological well-being [74]. Understanding players' psychological needs and experiences enables game designers to create games that are not only more engaging but also rewarding and pleasant for their intended audience. Figure A2 illustrates the main Quality Aspects of the PENS Questionnaire as defined by Vuorre et al. [75].

Appendix A3: Playful Experiences Questionnaire (PLEXQ). The PLEXQ is a seminal instrument crafted to evaluate a spectrum of playful encounters across diverse realms, spanning digital games, leisure endeavors, and social interactions [76]. Originating from the collaborative efforts of Sundstedt [77], the PLEXQ meticulously captures dimensions such as stimulation, pragmatic, momentary, and negative experiences, which serve as pivotal components in understanding the subjective essence of play [78].

Embracing a multidimensional perspective, the PLEXQ acknowledges the manifold manifestations of play within varying contexts [79]. Its design endeavors to furnish researchers, practitioners, and educators with a robust framework for assessing the multifaceted nature of playful engagements. Through an exploration of the distinct dimensions encapsulated by the PLEXQ, scholars can attain a nuanced comprehension of how play intersects with facets of well-being, creativity, social dynamics, and cognitive evolution across the lifespan [80].

The PLEXQ has found application in diverse research inquiries, probing the ramifications of gamification, interactive technologies, and recreational pursuits on individuals' playful experiences and overarching quality of life. Its utility extends beyond academia, offering insights that inform the design of interventions, educational programs, and recreational initiatives to foster enriching and fulfilling play experiences [81]. Figure A3 shows the PLEXQ.

Appendix A4: Player Experience Inventory (PXI). The PXI is a tool used to assess player experiences in video games, specifically focusing on both functional and psychosocial outcomes [82]. It was developed to capture key dimensions of player experience, addressing aspects such as enjoyment, challenge, and

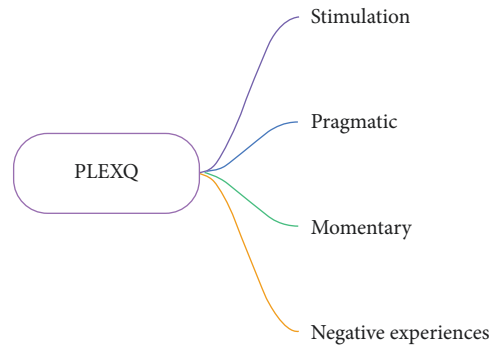

Figure A3: Playful Experiences Questionnaire (PLEXQ).

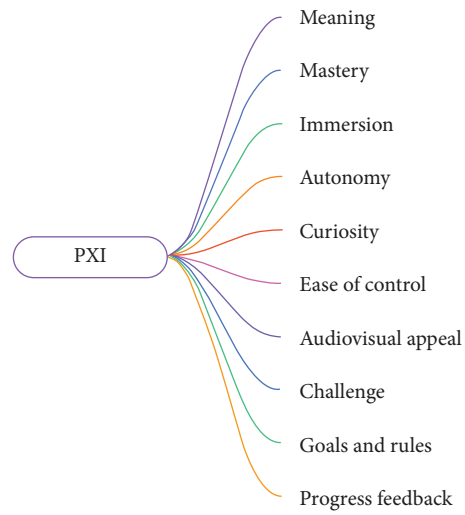

Figure A4: Player experience inventory (PXI).

engagement. It aims to measure how various game features and mechanics contribute to the overall player experience, linking these to psychological well-being and player satisfaction. The PXI helps researchers and game developers understand the deeper impacts of game design on players, beyond basic interaction, by evaluating the functional and emotional responses to gameplay [83].

In the context of this research paper, PXI's role is highlighted as part of the broader effort to create a user-centric quality model for GaaS. It is useful for evaluating how players interact with the game on a deeper level, ensuring that developers consider the full spectrum of player needs, such as emotional engagement, satisfaction, and immersion. This provides a comprehensive approach to assessing player experience, helping to guide game development towards meeting the psychological needs of players while maximizing player retention and engagement. Figure A4 shows the PLEXQ.

Appendix A5: The ITC Sense of Presence Inventory (ITC-SOPI). The ITC-SOPI is a seminal instrument used in VR and immersive technology research. It is specifically designed to quantify the subjective experience of presence within virtual environments. Developed by Lombard and Ditton, the ITC-SOPI comprises a series of carefully crafted items that aim to capture the extent to which users feel psychologically present

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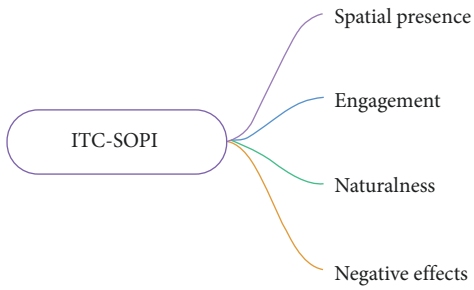

FIGURE A5: The ITC sense of presence inventory.

within a simulated reality. Presence, in this context, refers to the sensation of "being there" in the virtual environment, characterized by a sense of immersion, spatial presence, and involvement in the virtual experience [84].

The ITC-SOPI measures various dimensions of presence, including:

- Spatial presence: The feeling of being physically located within the virtual environment.
- Involvement: The degree of engagement and attention given to the virtual experience.
- Realism: The perception of the virtual environment as realistic and believable, which is crucial for creating a convincing virtual experience.

Figure A5 illustrates the main quality aspects of the ITC-SOPI as described by Masadeh et al. [81].

Data Availability Statement

The data are available upon request from the authors.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Ameen Shaheen: writing – review and editing, writing – original draft, visualization, validation, supervision, resources, project administration, methodology, investigation, formal analysis, conceptualization. **Ahmad Alkhatib:** writing – review and editing, writing – original draft, visualization, validation, formal analysis, conceptualization. **Mahmoud Farfoura:** writing – review and editing, writing – original draft, visualization, validation. **Rand Albustanji:** writing – review and editing, writing – original draft, validation, formal analysis.

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